DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

Cyber Security | Continuous Assessment II | Case Study (Experiential Learning)

Phishing Email Detection using Machine Learning and NLP: A Case Study of the US$100 Million Google–Facebook Business Email Compromise Fraud (2013–2015)

Group No.: _____

Topic: Phishing Email Detection using ML/NLP (Unit 4)

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Abstract

Business Email Compromise (BEC) is a form of targeted phishing in which an attacker impersonates a trusted party to obtain a payment. Between 2013 and 2015, a Lithuanian national, Evaldas Rimašauskas, used this technique to obtain roughly US$121 million from Google and Facebook by posing as Quanta Computer, a Taiwanese hardware supplier that worked with both companies [1]–[4]. The emails contained no malware and no malicious links, so conventional filters had nothing to block. This report studies the case in detail and then examines how machine learning (ML) and natural language processing (NLP) can be used to detect such emails. We trained five classifiers on the public CEAS-2008 corpus of 39,154 labelled emails using TF-IDF text features and eight engineered metadata features. A linear Support Vector Machine achieved the best result, with an F1-score of 0.9984 and a false-positive rate of 0.20% on a held-out test set of 7,831 emails. We then tested the same models on a synthetic set of 60 BEC-style invoice emails and 60 legitimate invoice emails modelled on the case. The generic models were unstable on this business-payment data: one detected none of the BEC emails, and others wrongly flagged up to 100% of genuine invoices. A small model built on BEC-specific signals (look-alike sender domain, bank-change wording, bank-country mismatch and urgency) separated the two classes completely on this synthetic set. We conclude that text classifiers trained on generic phishing data are not sufficient for BEC, and that detection must combine language features with sender authentication, vendor context and a mandatory out-of-band verification process.

Keywords: Business Email Compromise, phishing detection, machine learning, natural language processing, TF-IDF, Support Vector Machine, social engineering, invoice fraud, email authentication

1. Introduction

Phishing is the use of deceptive messages to make a person reveal information or take an action that benefits an attacker. Most people associate phishing with mass emails carrying suspicious links. Business Email Compromise is a more targeted variant. The attacker studies an organisation, impersonates a supplier, executive or partner, and asks an employee to pay an invoice or change payment details. Because the request fits into normal business activity, it is often paid without question.

BEC is now one of the costliest forms of cybercrime. The FBI's Internet Crime Complaint Center (IC3) recorded adjusted BEC losses of about US$360 million in 2016, rising to about US$2.9 billion in 2023 [5]–[8], and reported global exposed losses of more than US$55 billion between October 2013 and December 2023 [10]. These amounts are far larger than losses from ransomware reported to the same body over the same years [8], [9].

The Google–Facebook fraud is one of the clearest examples of BEC on record. It shows that technical sophistication on the defending side does not protect against an attack that targets business processes and human trust. It is also a useful reference case for detection research, because the emails involved would have passed most traditional security checks.

This report has two aims. The first is to give a complete account of the case: who was involved, how the attack was carried out, how it was discovered and prosecuted, and why it succeeded. The second is to design and evaluate an ML/NLP approach for detecting phishing and BEC emails, and to test how well such models handle emails similar to those used in the case. Section

2 reviews related work. Section 3 describes the case and our methodology. Section 4 gives implementation details. Section 5 presents and discusses results, and Section 6 concludes.

2. Literature Review

2.1 The Business Email Compromise threat

The FBI classifies BEC as a scam that targets businesses and individuals who make wire transfers, typically by compromising or spoofing legitimate email accounts [10]. IC3 annual reports show that BEC has consistently been among the top categories by financial loss since at least 2016 [5]–[9]. The Verizon Data Breach Investigations Report similarly identifies pretexting, the core technique of BEC, as a leading social-engineering pattern in confirmed incidents [11]. Mitnick and Simon describe how attackers exploit trust, authority and routine to bypass technical controls [36], and Cialdini's principles of influence, especially authority and scarcity, explain why urgent requests from apparent partners are obeyed [35].

2.2 Classical machine-learning approaches

Early work treated phishing detection as a text-classification problem. Fette, Sadeh and Tomasic proposed PILFER, which used ten features such as IP-based URLs, link-text mismatch and the presence of JavaScript, and reached high accuracy with a random forest [12]. Abu-Nimeh et al. compared logistic regression, classification and regression trees, Bayesian additive regression trees, support vector machines, random forests and neural networks on a phishing corpus and found that no single model dominated on all metrics [13]. Bergholz et al. added model-based features derived from email content and reported further improvements [14]. Surveys by Almomani et al. [19], Khonji et al. [20] and Karim et al. [37] summarise this line of research and note that most features rely on URLs, attachments and spam-like content. These features are of limited use against BEC, where the email usually contains neither.

2.3 Text representation and deep learning

TF-IDF weighting, introduced for information retrieval [28], remains a strong baseline for email classification when combined with linear models such as logistic regression or support vector machines [26]. Word embeddings [38], convolutional networks for sentences [39] and recurrent networks such as LSTM [24] allow models to capture word order and meaning. Transformer models [23], and in particular BERT [21] and its smaller variant DistilBERT [22], have produced state-of-the-art results on many text-classification tasks. A systematic review by Salloum et al. found that NLP-based phishing detectors increasingly use such contextual representations, while also noting the lack of up-to-date public datasets and the dominance of older corpora [18].

2.4 Detection of targeted and BEC attacks

Targeted attacks require different features. Ho et al. detected credential spear-phishing in a large enterprise by combining the content of an email with features describing the sender's reputation and history [17], and later characterised lateral phishing sent from compromised internal accounts [16]. Cidon et al. presented BEC-Guard, which detects BEC using both the content of the email and metadata about the relationship between the sender and recipient, and reported high precision on real enterprise traffic [15]. A common conclusion across these studies is that the content of a BEC email alone is often indistinguishable from a legitimate

message. Detection must therefore also consider who the sender is, how they relate to the recipient and whether the request departs from normal behaviour.

2.5 Email authentication standards

Sender Policy Framework (SPF) [32], DomainKeys Identified Mail (DKIM) [33] and Domain-based Message Authentication, Reporting and Conformance (DMARC) [31] allow a receiving server to check whether an email was sent by a server authorised by the claimed domain. They prevent exact-domain spoofing when properly enforced, but do not stop an attacker who registers a look-alike domain and authenticates it correctly. Edit-distance measures such as the Levenshtein distance [34] are commonly used to detect such look-alike domains.

2.6 Research gap

Most published detectors are trained and evaluated on public corpora such as Enron [29], the Nazario phishing corpus [40], SpamAssassin [41] and CEAS-2008 [30]. These contain mass phishing and spam rather than BEC. Reported accuracies above 0.95 on these datasets [18] therefore say little about performance on BEC. This report tests that gap directly by applying models trained on CEAS-2008 to BEC-style emails modelled on the Google–Facebook case.

3. Methodology

Our methodology has two parts. The first is a case analysis, built from primary legal sources and contemporary reporting. The second is an experimental study of ML/NLP detection.

3.1 Case-analysis method

The case was reconstructed from the U.S. Department of Justice (DOJ) press releases for the Southern District of New York (SDNY) on the charges [1], the guilty plea [2] and the sentence [3], from the Fortune report that first named the victims [4], and from FBI IC3 data for context [5]–[10]. The attack was then broken down into stages, and each stage was mapped to the control that failed and to a signal that a detection system could use.

3.2 The case in brief

Around 2013, Rimašauskas registered a company in Latvia with the same name as Quanta Computer, a real Taiwanese manufacturer that supplied hardware to both victims, and opened bank accounts in its name in Latvia and Cyprus [1], [2]. He then sent emails to employees and agents of the two companies from accounts that appeared to belong to Quanta staff. These emails asked for payment of invoices for goods and services that had apparently been provided. Between 2013 and 2015 the victims transferred over US$100 million to the attacker's accounts; the DOJ later put the total at more than US$120 million, approximately US$98 million from Facebook and US$23 million from Google [3], [4]. To persuade banks that the transfers were genuine, he provided forged invoices, contracts and letters bearing forged signatures and corporate stamps [1]. The funds were rapidly moved to accounts in other countries, including Slovakia, Lithuania, Hungary and Hong Kong [1]. Fig. 1 summarises the attack chain and Table 1 the timeline.

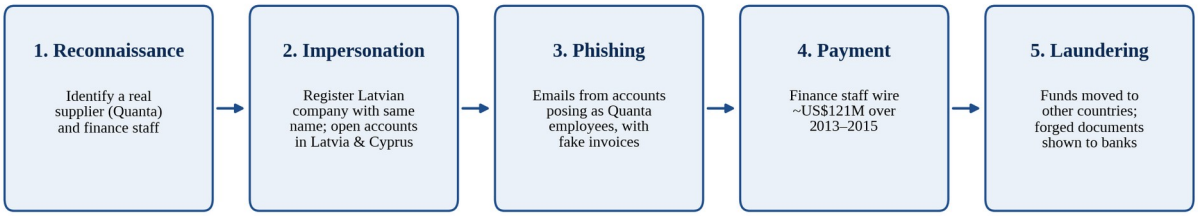


Fig. 1. Attack chain of the Google–Facebook BEC fraud (compiled from [1]–[4])

Table 1. Timeline of the case

Date	Event
c. 2013	Latvian company named “Quanta Computer” registered; bank accounts opened in Latvia and Cyprus [1]
2013–2015	Phishing emails and fake invoices sent; victims wire over US$100 million [1], [3]
2015	Scheme ends; victims detect the fraud and begin recovering funds [4]
March 2017	Rimašauskas arrested in Lithuania; DOJ announces charges without naming the victims [1]
April 2017	Fortune identifies the victims as Google and Facebook [4]
August 2017	Extradited to the United States [2]
March 2019	Pleads guilty to one count of wire fraud; agrees to forfeit US$49.7 million [2]
December 2019	Sentenced to five years in prison [3]

The charges included wire fraud, three counts of money laundering and aggravated identity theft [1]. Both companies stated that they had recovered most or all of the money [4]. The analysis identified four reasons for the attack's success: it exploited a real supplier relationship, so large invoices looked routine; the emails carried no malicious payload; forged documents satisfied bank checks; and there was no mandatory independent verification of payment requests from the supplier. These findings shaped the features used in the experiments.

3.3 Detection pipeline

Fig. 2 shows the detection pipeline. It has five stages: data collection, pre-processing, feature extraction, classification and decision. Instead of simply deleting suspicious emails, the final stage assigns a risk score, and high-risk emails that request payment are held until the request is confirmed through an independent channel such as a phone call to a known contact.

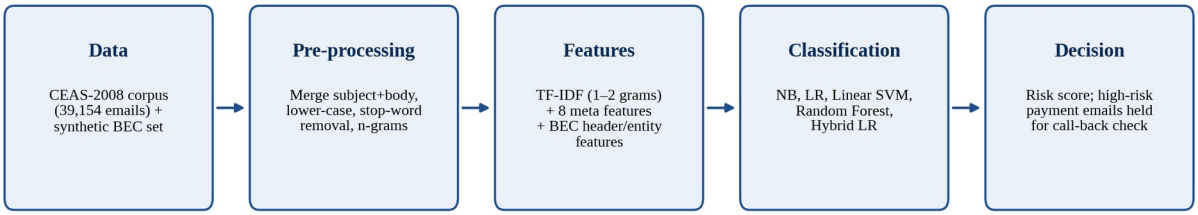


Fig. 2. Proposed ML/NLP detection pipeline

3.4 Datasets

CEAS-2008. The main dataset is the CEAS-2008 corpus from the Conference on Email and Anti-Spam live challenge [30], in the cleaned form that includes sender, receiver, date, subject, body, a URL indicator and a binary label. It contains 39,154 emails: 21,842 labelled as phishing or spam (label 1) and 17,312 labelled as legitimate (label 0). It was split into 31,323 training and 7,831 test emails (80/20, stratified by label, random seed 42).

Synthetic BEC set. Public BEC emails are very rare because organisations do not release them. We therefore created 120 synthetic emails based on the patterns described in the case. Sixty are legitimate invoices from six fictitious vendors, sent from the vendor's correct domain and paid to the usual account. Sixty are BEC emails sent from a look-alike version of a vendor domain (for example, a letter replaced by a similar one, or an extra suffix), requesting payment to a new account in a country unrelated to the vendor, often with urgency. Both classes use the same invoice vocabulary, amounts and structure, so that differences come only from BEC-specific signals. This set is used only for testing and is clearly smaller and simpler than real traffic.

3.5 Features

Three groups of features were used, as summarised in Table 2.

Table 2. Feature groups

Group	Features	Rationale
Text (NLP)	TF-IDF of subject + body, unigrams and bigrams, 50,000 terms, English stop words removed, sublinear term frequency	Captures phishing vocabulary and phrasing [28]
Metadata	URL present, log body length, exclamation ratio, upper-case ratio, urgency-term rate, payment-term rate, bank-change-term rate, money amount present	Captures style and intent independent of exact words
BEC-specific	Sender domain is not a known vendor but close to one (Levenshtein distance); bank-change wording; bank country differs from vendor country; urgency wording	Directly encodes the signals of the Quanta impersonation [15], [34]

3.6 Models

Five models were trained on CEAS-2008: Multinomial Naive Bayes on TF-IDF features; Logistic Regression on TF-IDF features; a linear Support Vector Machine on TF-IDF features [26]; a Random Forest [25] on the eight metadata features only; and a hybrid Logistic Regression on TF-IDF and metadata features together. These were chosen because they are well established, fast to train and interpretable, which matters in a security setting where analysts must understand why an email was flagged. A sixth model, a Logistic Regression on the four BEC-specific features, was trained and evaluated on the synthetic set using 5-fold stratified cross-validation, because those features require vendor information that CEAS-2008 does not contain.

3.7 Evaluation

Models are compared on accuracy, precision, recall, F1-score and false-positive rate (the share of legitimate emails wrongly flagged). For payment-related emails, recall is the most important metric, because a missed BEC email can cost millions while a false alarm costs one verification call. We also report 5-fold cross-validated F1 on the training data for the hybrid model. On the synthetic set, we report the BEC detection rate and the rate at which legitimate invoices are wrongly flagged.

4. Implementation Details

The system was implemented in Python 3 using scikit-learn [27], pandas, NumPy, SciPy and Matplotlib. All experiments ran on a standard CPU without a GPU. The main steps were as follows.

1. **Loading and cleaning.** The CEAS-2008 CSV file was loaded with pandas. Rows with an empty body were removed and duplicate subject–body pairs dropped (none were found). Missing subjects were replaced with an empty string.
2. **Text preparation.** Subject and body were concatenated and converted to lower case. TfidfVectorizer was configured with ngram_range=(1, 2), max_features=50000, min_df=2, stop_words="english" and sublinear_tf=True. The vectoriser was fitted on the training split only, to avoid information from the test set leaking into the model.
3. **Metadata features.** A function computed the eight metadata features for each email. Keyword-rate features were normalised per 100 words. All metadata features were scaled with MaxAbsScaler and combined with the sparse TF-IDF matrix using scipy.sparse.hstack.
4. **Model training.** MultinomialNB(alpha=0.1), LogisticRegression(C=10, max_iter=2000), LinearSVC(C=1.0) and RandomForestClassifier(n_estimators=300) were trained on the training split. Training time was recorded for each.
5. **BEC features.** For the synthetic set, the sender domain was extracted from the From field and compared with a list of known vendor domains using a Levenshtein-distance function. A domain that was not on the list but close to a listed one was marked as a look-alike. Simple keyword rules marked bank-change and urgency wording, and the bank country stated in the email was compared with the vendor's country.
6. **Evaluation and plots.** Metrics were computed with sklearn.metrics, and figures were produced with Matplotlib. A fixed random seed (42) was used for the split, the cross-validation folds and the Random Forest, so the results can be reproduced.

A condensed version of the core training code is shown below.

```
tf = TfidfVectorizer(ngram_range=(1,2), max_features=50000, min_df=2,
                     stop_words='english', sublinear_tf=True)
X_tr = tf.fit_transform(text_train); X_te = tf.transform(text_test)
svm = LinearSVC(C=1.0).fit(X_tr, y_train)
pred = svm.predict(X_te)
print(f1_score(y_test, pred), recall_score(y_test, pred))
```

5. Results and Discussion

5.1 Performance on CEAS-2008

Table 3 shows the test-set results for the five models trained on CEAS-2008, and Fig. 3 compares precision, recall and F1-score.

Table 3. Results on the CEAS-2008 test set (7,831 emails)

Model	Accuracy	Precision	Recall	F1	FPR	Train (s)
Naive Bayes (TF-IDF)	0.9890	0.9991	0.9812	0.9901	0.12%	0.02
Logistic Regression (TF-IDF)	0.9971	0.9975	0.9973	0.9974	0.32%	0.18
Linear SVM (TF-IDF)	0.9982	0.9984	0.9984	0.9984	0.20%	0.25
Random Forest (metadata only)	0.9588	0.9602	0.9661	0.9631	5.05%	4.66
Logistic Regression (hybrid)	0.9972	0.9975	0.9975	0.9975	0.32%	0.30

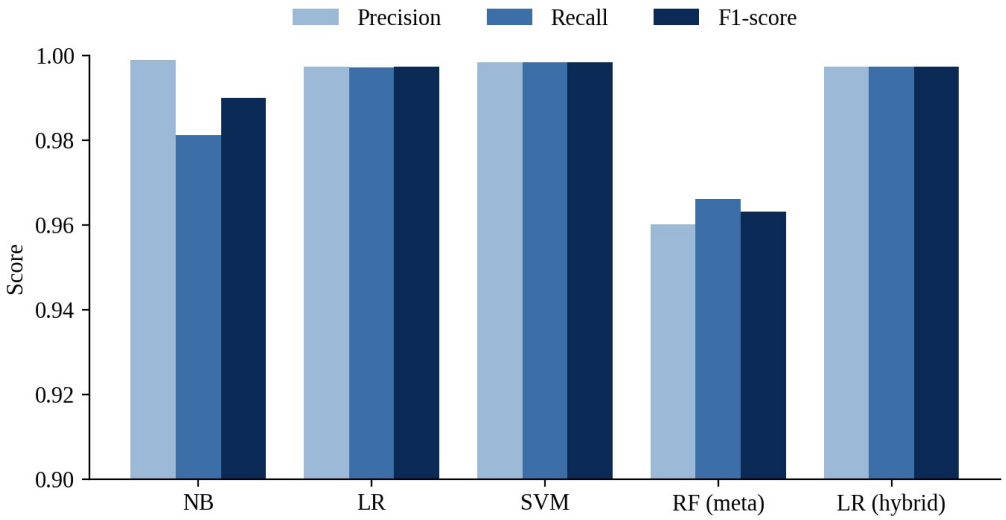


Fig. 3. Precision, recall and F1-score of each model on CEAS-2008

The linear SVM gave the best overall result, with an F1-score of 0.9984 and only 7 legitimate emails wrongly flagged out of 3,462. Fig. 4 shows its confusion matrix. Logistic Regression and the hybrid Logistic Regression performed almost identically (F1 0.9974 and 0.9975). The 5-fold cross-validated F1 of the hybrid model on the training data was 0.9958 ± 0.0007, which shows that the result is stable and not due to a lucky split. Naive Bayes had the highest precision but lower recall: it missed 82 phishing emails. The Random Forest using only eight metadata features reached an F1 of 0.9631, showing that style and intent features carry useful information on their own, but with a much higher false-positive rate (5.05%) than the text models.

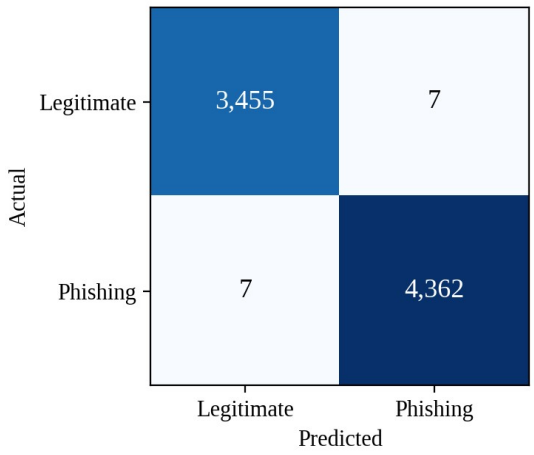


Fig. 4. Confusion matrix of the linear SVM on the CEAS-2008 test set

These results are in line with published work that reports accuracies above 0.95 on public corpora [12], [13], [18]. However, CEAS-2008 is dominated by mass spam and phishing with distinctive vocabulary and links, which makes the task relatively easy.

5.2 Stress test on BEC-style emails

Table 4 and Fig. 5 show what happened when the CEAS-trained models were applied to the synthetic BEC set, alongside the BEC-specific model.

Table 4. Results on the synthetic BEC set (60 BEC and 60 legitimate invoice emails)

Model	BEC emails detected	Legitimate invoices wrongly flagged
Naive Bayes (TF-IDF)	0.0%	0.0%
Logistic Regression (TF-IDF)	100.0%	41.7%
Linear SVM (TF-IDF)	100.0%	3.3%
Random Forest (metadata only)	45.0%	100.0%
Logistic Regression (hybrid)	88.3%	98.3%
BEC-specific features + LR (5-fold CV)	100.0%	0.0%

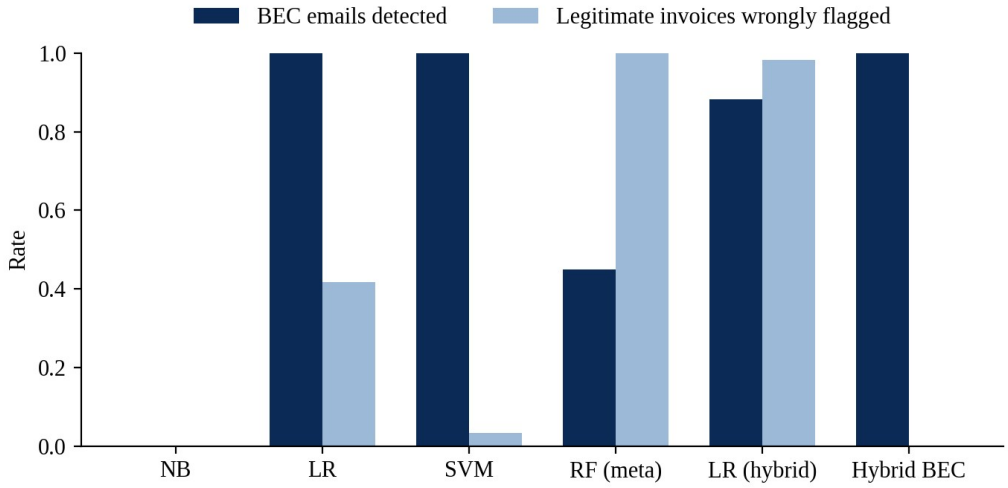


Fig. 5. BEC detection rate and false-flag rate on the synthetic BEC set

The generic models behaved very differently from one another on business-payment emails, even though all of them scored above 0.96 F1 on CEAS-2008. Naive Bayes detected none of the BEC emails, so in a real deployment every fraudulent invoice would have reached the finance team. The Logistic Regression models and the metadata Random Forest detected many or all BEC emails but also flagged a large share of genuine invoices (41.7% to 100.0%). In other words, they reacted to invoice and payment vocabulary in general rather than to what makes an invoice fraudulent. A filter that blocks most real supplier invoices would be switched off in practice. The linear SVM performed well here (100.0% detected, 3.3% false flags), but this should be treated with caution, because the synthetic emails follow a small number of templates and the bank-change sentences contain words that also appear in CEAS phishing.

The model using only four BEC-specific signals separated the two classes completely (100% detected, 0% false flags). This result is optimistic, since the synthetic data was designed around these signals. Its value is in showing *which* information resolves the ambiguity. The fraudulent and genuine invoices in the synthetic set differ only in the sender domain, the destination of the money and the urgency of the request, exactly as in the Quanta case. A model that knows the organisation's real vendors, their domains and their usual bank countries can separate them; a model that sees only the text cannot reliably do so.

5.3 Application to the Google–Facebook case

Applied to the case, the signals in Table 2 map directly onto the attack. An email from a domain resembling Quanta's but not authenticated by Quanta's real domain would fail the look-alike and DMARC checks [31]. A request to pay a Taiwanese supplier into accounts in Latvia or Cyprus would trigger the country-mismatch feature. A first payment to a new account for an existing vendor would trigger a behavioural alert. None of these checks needs malware or a malicious link to be present. However, the attacker also controlled a company with the same name as the real supplier, and bank checks were satisfied with forged documents [1]. This shows that detection should feed into a process step: any change in a supplier's payment details must be confirmed by phone using contact details already on file, not details supplied in the email.

5.4 Limitations

- The synthetic BEC set is small and template-based. Real BEC emails are more varied, so the BEC results show a trend, not a reliable performance estimate.
- CEAS-2008 dates from 2008 and does not reflect current phishing language or the use of generative AI by attackers.
- Header authentication results (SPF, DKIM and DMARC) were not available in the dataset and were not evaluated directly.
- Transformer models such as DistilBERT [22] were not trained, owing to computing limits; they are left for future work.

6. Conclusion

The Google–Facebook fraud showed that one person, without any malicious code, could obtain over US$100 million from two of the world's most technically capable companies by exploiting

a real supplier relationship and weak payment verification. Our experiments show that classical ML models trained on a public phishing corpus can detect mass phishing extremely well, with the linear SVM reaching an F1-score of 0.9984. They also show that this success does not transfer reliably to BEC: on business-payment emails, the same family of models either missed every fraudulent invoice or flagged a large share of genuine ones. Features that describe the sender's identity, the destination of the money and the departure from a vendor's normal behaviour were what separated fraudulent invoices from real ones.

We therefore recommend a layered defence: enforced email authentication with look-alike domain detection; ML/NLP models that combine text with vendor-aware metadata to score payment-related emails; mandatory out-of-band verification of any change in payment details, with dual approval above a value threshold; and regular training of finance staff on BEC warning signs. Future work should evaluate transformer models such as DistilBERT, use real anonymised BEC samples where they can be obtained, and include header authentication results as features.

References

[1] U.S. Department of Justice, U.S. Attorney's Office, Southern District of New York, "Lithuanian man arrested for theft of over $100 million in fraudulent email compromise scheme against multinational internet companies," press release, Mar. 21, 2017.

[2] U.S. Department of Justice, U.S. Attorney's Office, Southern District of New York, "Lithuanian man pleads guilty to wire fraud for theft of over $100 million in fraudulent business email compromise scheme against multinational internet companies," press release, Mar. 20, 2019.

[3] U.S. Department of Justice, U.S. Attorney's Office, Southern District of New York, "Lithuanian man sentenced to 5 years in prison for theft of over $120 million in fraudulent business email compromise scheme," press release, Dec. 2019.

[4] J. J. Roberts, "Exclusive: Facebook and Google were victims of $100M payment scam," Fortune, Apr. 27, 2017.

[5] Federal Bureau of Investigation, Internet Crime Complaint Center, "2016 Internet Crime Report," 2017.

[6] Federal Bureau of Investigation, Internet Crime Complaint Center, "2019 Internet Crime Report," 2020.

[7] Federal Bureau of Investigation, Internet Crime Complaint Center, "2022 Internet Crime Report," 2023.

[8] Federal Bureau of Investigation, Internet Crime Complaint Center, "2023 Internet Crime Report," 2024.

[9] Federal Bureau of Investigation, Internet Crime Complaint Center, "2024 Internet Crime Report," 2025.

[10] Federal Bureau of Investigation, Internet Crime Complaint Center, "Business Email Compromise: The $55 Billion Scam," Public Service Announcement I-091124-PSA, Sep. 2024.

[11] Verizon, "2024 Data Breach Investigations Report," Verizon Business, 2024.

[12] I. Fette, N. Sadeh, and A. Tomasic, "Learning to detect phishing emails," in Proc. 16th Int. Conf. World Wide Web (WWW), 2007, pp. 649–656.

[13] S. Abu-Nimeh, D. Nappa, X. Wang, and S. Nair, "A comparison of machine learning techniques for phishing detection," in Proc. Anti-Phishing Working Group eCrime Researchers Summit, 2007, pp. 60–69.

[14] A. Bergholz, J. De Beer, S. Glahn, M.-F. Moens, G. Paaß, and S. Strobel, "New filtering approaches for phishing email," Journal of Computer Security, vol. 18, no. 1, pp. 7–35, 2010.

[15] A. Cidon, L. Gavish, I. Bleier, N. Korshun, M. Schweighauser, and A. Tsitkin, "High precision detection of business email compromise," in Proc. 28th USENIX Security Symposium, 2019, pp. 1291–1307.

[16] G. Ho, A. Cidon, L. Gavish, M. Schweighauser, V. Paxson, S. Savage, G. M. Voelker, and D. Wagner, "Detecting and characterizing lateral phishing at scale," in Proc. 28th USENIX Security Symposium, 2019, pp. 1273–1290.

[17] G. Ho, A. Sharma, M. Javed, V. Paxson, and D. Wagner, "Detecting credential spearphishing in enterprise settings," in Proc. 26th USENIX Security Symposium, 2017, pp. 469–485.

[18] S. Salloum, T. Gaber, S. Vadera, and K. Shaalan, "A systematic literature review on phishing email detection using natural language processing techniques," IEEE Access, vol. 10, pp. 65703–65727, 2022.

[19] A. Almomani, B. B. Gupta, S. Atawneh, A. Meulenberg, and E. Almomani, "A survey of phishing email filtering techniques," IEEE Communications Surveys & Tutorials, vol. 15, no. 4, pp. 2070–2090, 2013.

[20] M. Khonji, Y. Iraqi, and A. Jones, "Phishing detection: A literature survey," IEEE Communications Surveys & Tutorials, vol. 15, no. 4, pp. 2091–2121, 2013.

[21] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[22] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," arXiv:1910.01108, 2019.

[23] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 5998–6008.

[24] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[25] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[26] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[27] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[28] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Information Processing & Management, vol. 24, no. 5, pp. 513–523, 1988.

[29] B. Klimt and Y. Yang, "The Enron corpus: A new dataset for email classification research," in Proc. European Conf. Machine Learning (ECML), 2004, pp. 217–226.

[30] CEAS 2008 Live Spam Challenge corpus, Conference on Email and Anti-Spam (CEAS), 2008; cleaned version as compiled in A. Al-Subaiey et al., "Novel interpretable and robust web-based AI platform for phishing email detection," Computers and Electrical Engineering, 2024.

[31] M. Kucherawy and E. Zwicky, Eds., "Domain-based Message Authentication, Reporting, and Conformance (DMARC)," RFC 7489, IETF, 2015.

[32] S. Kitterman, "Sender Policy Framework (SPF) for authorizing use of domains in email, version 1," RFC 7208, IETF, 2014.

[33] D. Crocker, T. Hansen, and M. Kucherawy, Eds., "DomainKeys Identified Mail (DKIM) signatures," RFC 6376, IETF, 2011.

[34] V. I. Levenshtein, "Binary codes capable of correcting deletions, insertions, and reversals," Soviet Physics Doklady, vol. 10, no. 8, pp. 707–710, 1966.

[35] R. B. Cialdini, Influence: The Psychology of Persuasion, rev. ed. New York, NY, USA: Harper Business, 2006.

[36] K. D. Mitnick and W. L. Simon, The Art of Deception: Controlling the Human Element of Security. Indianapolis, IN, USA: Wiley, 2002.

[37] A. Karim, S. Azam, B. Shanmugam, K. Kannoorpatti, and M. Alazab, "A comprehensive survey for intelligent spam email detection," IEEE Access, vol. 7, pp. 168261–168295, 2019.

[38] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," arXiv:1301.3781, 2013.

[39] Y. Kim, "Convolutional neural networks for sentence classification," in Proc. EMNLP, 2014, pp. 1746–1751.

[40] J. Nazario, "Phishing corpus," monkey.org, 2005–2007. [Online].

[41] Apache SpamAssassin Project, "SpamAssassin public mail corpus," 2002–2005. [Online].